

Free particle evolution

Consider the Schrodinger Equation

$$-\frac{\hbar^2}{2m} \frac{\partial^2 \psi(x,t)}{\partial x^2} + V(x,t) \psi(x,t) = i\hbar \frac{\partial \psi(x,t)}{\partial t}$$

For a free particle we may set $V(x,t) = 0$

$$-\frac{\hbar^2}{2m} \frac{\partial^2 \psi(x,t)}{\partial x^2} = i\hbar \frac{\partial \psi(x,t)}{\partial t}$$

From separation of variables $\psi(x,t) = \phi(x) \varphi(t)$

$$-\frac{\hbar^2}{2m} \frac{\phi''(x)}{\phi(x)} = i\hbar \frac{\varphi'(t)}{\varphi(t)} = E$$

$$\text{hence, } \phi''(x) + \frac{2mE}{\hbar^2} \phi(x) = 0$$

$$\Rightarrow \phi(x) = e^{i\sqrt{\frac{2mE}{\hbar^2}} x}$$

Postulate; $E = \frac{p^2}{2m} \leftarrow$ free particle in classical mechanics,

$$\Rightarrow \phi(x) = e^{i\frac{p}{\hbar} x}$$

$$\text{Now, } \frac{\varphi'(t)}{\varphi(t)} = -\frac{iE}{\hbar}$$

$$\text{Solution, } \varphi(t) = e^{-iEt/\hbar} = e^{-i\frac{p^2}{2m\hbar} t}$$

Hence, $\psi(x,t) = e^{i(\frac{p}{\hbar} x - \frac{p^2}{2m\hbar} t)} \leftarrow$ Most general solution.

Since the ψ - E is linear,

Any linear combination is also a solution,

$$\psi(x,t) = \sum \phi(p) e^{i(\frac{p}{\hbar} x - \frac{p^2}{2m\hbar} t)}$$

these constants are the constants that appear in the Fourier series that depend on the momentum, also called the momentum space at $t=0$.

These $\phi(p)$ can be obtained by the inverse Fourier transform,

$$\phi(p) = \frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^{\infty} \psi(x, 0) e^{-ipx/\hbar} dx = \frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^{\infty} \left(\sum \phi(p) e^{i(\frac{p}{\hbar}x)} \right) e^{-\frac{ipx}{\hbar}} dx$$

Example:

Gaussian wave packet,

Consider a wavefunction whose profile at $t=0$ is,

$$\psi(x, 0) = e^{i\frac{p_0}{\hbar}x} e^{-\frac{x^2}{2\sigma^2}}$$

The momentum wavefunction is

$$\begin{aligned} \phi(p) &= \int_{-\infty}^{\infty} \psi(x, 0) e^{-ipx/\hbar} dx = \int_{-\infty}^{\infty} e^{ip_0 x/\hbar} e^{-\frac{x^2}{2\sigma^2}} dx \\ &= e^{-(p-p_0)^2/2(\hbar/\sigma)^2} \end{aligned}$$

We have,

$$\psi(x, 0) \psi(x, 0)^* = e^{-\frac{x^2}{2(\sigma_x^2)}} \quad / \quad \phi(p, 0) \phi(p, 0)^* = e^{-\frac{(p-p_0)^2}{2(\frac{\hbar^2}{2\sigma^2})}}$$

$\uparrow$ $\uparrow$
 σ_x^2 σ_p^2

Hence,

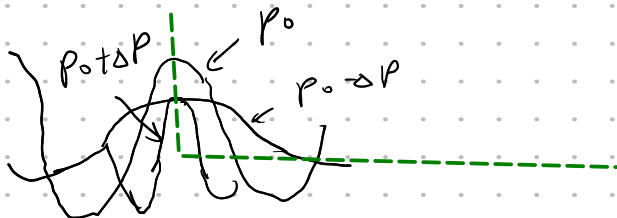
$$\sigma_x \sigma_p = \frac{\sigma}{\sqrt{2}} \frac{\hbar}{\sqrt{2} \sigma} = \frac{\hbar}{2} \leftarrow \text{Hence it achieves minimum uncertainty, it is also called minimum uncertainty wave packet.}$$

Hence we may write the time evolution of the wave packet as,

$$\psi(x, t) = \sum_p e^{-(p-p_0)^2/2(\hbar/\sigma)^2} e^{i(\frac{p}{\hbar}x - \frac{p^2}{2m\hbar}t)}$$

This solution may be regarded as a sum of waves $e^{i(\frac{p}{\hbar}x - \frac{p^2}{2m\hbar}t)}$ each modulated by a factor of $e^{-(p-p_0)^2/2(\hbar^2/\sigma^2)}$, with max at $p=p_0$.

Picture,



As time evolves,

the wave function moves with group velocity

$\frac{\partial \omega(k)}{\partial k} \rightarrow$ where we have

$$\psi(x, t) = \sum_p \phi(p) e^{i(\frac{p}{\hbar}x - \frac{p^2}{2m\hbar}t)} = \sum_p \phi(p) e^{i(\frac{p}{\hbar}x - \omega(p)t)}$$

And,

$$\frac{\partial \omega}{\partial k} = \frac{\partial \omega}{\partial p} \frac{\partial p}{\partial k} = \frac{\partial \omega}{\partial p} \frac{\partial (\hbar k)}{\partial k} = \hbar \frac{\partial \omega(p)}{\partial p}$$

$$\frac{\partial \omega}{\partial k} = \hbar \frac{\partial}{\partial p} \left(\frac{p^2}{2m\hbar} \right) = \frac{p}{m} \leftarrow \text{velocity of the wave with}$$

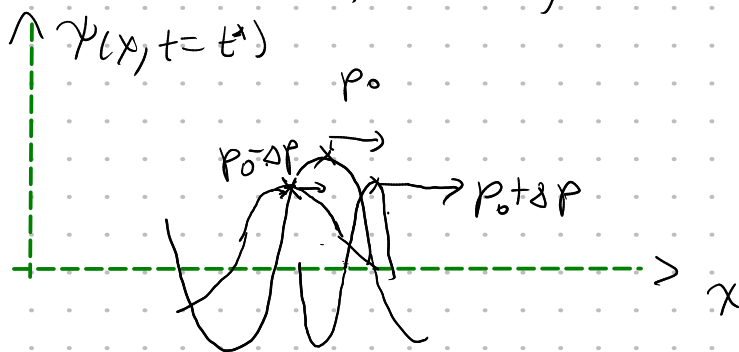
momentum p , since all the waves are modulated by

$e^{-(p-p_0)^2/2(\hbar^2/2\sigma^2)} \leftarrow$ on average the group velocity would be,

$$\frac{p_0}{m} = v \leftarrow \text{this is the classical velocity.}$$

The spread of the wavefunction would be due to some waves stay ahead while other stay behind,

Pictorially,



Furthermore, the probability density $|\psi(x, t)|^2$ is given by,

$$|\psi(x, t)|^2 = \frac{\pi}{\sqrt{\sigma_0^4 + \hbar^2 t^2 / 2m^2}} e^{-(x - \frac{p_0}{m}t)^2 / 2(\sigma_0^2 + \frac{\hbar^2 t^2}{4m\sigma_0^2})}$$

At $t=0$ the wave has a spread of σ_0

Let's estimate the time t^* at which the spread doubles,

$$4\sigma_0^2 = \sigma_0^2 + \frac{\hbar^2 t^2}{4m^2 \sigma_0^2}$$

$$t^* = \frac{12\sigma_0^4 (4m^2)}{\hbar^2}$$

Hence, we expect that for particles of large mass there is not much spread of the wave function.

References:

Sakurai, (Jun John) Sakurai, and Jim J. Napolitano. Modern Quantum Mechanics. Pearson, 2014.